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No spill straw

Abstract

Aspects disclosed herein generally relate to beverage systems. Particularly, beverage systems comprising a straw mechanism that is removably secured to a lid structure. The beverage systems in accordance with aspects herein have straw mechanisms that are selectively leak-proof.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 18/516,771, filed on Nov. 21, 2023 entitled “No Spill Straw” and claims the benefit of U.S. Provisional Application No. 63/529,125 entitled “No Spill Straw,” filed on Jul. 26, 2023 and U.S. Provisional Application No. 63/529,988 entitled “No Spill Straw,” filed on Jul. 31, 2023.

TECHNICAL FIELD

(1) Aspects provided herein relate to portable beverage accessories, such as insulated travel cups, shaker cups, and the like.

BACKGROUND

(2) Travel beverage accessories often include insulated cups and non-insulated cups, double wall cups, or any other suitable beverage container that includes a fitted lid to prevent accidental spills. The lids can be simple screw on lids, or press-on lids, and may not always be leak-proof. Lids that are supposedly leak-proof oftentimes include a mechanism that typically requires two hands to operate, and can thus be cumbersome to use.

SUMMARY

(3) At a high level, a beverage container may include a lid structure coupled to a base structure to define an interior volume for holding and/or dispensing a beverage held within the beverage container. The base structure may be a cup shaped container that may optionally be insulating to keep drinks cold/hot for extended periods of time (e.g., between 4 hours and 48 hours). The lid structure may have a spout portion or a straw mechanism for dispensing the contents of the beverage container when the lid structure and the base structure are engaged with one another. In order to provide a leak-proof beverage container, the straw mechanism is able to rotatably transition between a closed state (i.e., a not dispensing state, a no-spill state, a traveling state, or a not in use) and an open state (i.e., a dispensing state). When in its closed state, the straw mechanism prevents the contents of the beverage container from spilling out or otherwise being dispensed or consumed from the beverage container. When in its open state, the straw mechanism allows the contents of the beverage container to be consumed or otherwise dispensed from the beverage container.

(4) The lid structure in accordance with aspects herein comprises a straw mechanism and a body portion. The body portion has a threaded engagement section configured to securely engage the base structure of the beverage container, and an engagement cavity configured to receive and engage with the straw mechanism, as will become more apparent in the description with respect to the figures provided below. The straw mechanism and the body portion are disengageably engaged to each other to facilitate cleaning of the lid structure. Further, the straw mechanism and the body portion are rotatably engaged to provide a leak-proof lid structure for traveling or general portability. This is advantageous over conventional straw-based beverage containers, where the straw extends through a penetrable opening in the lid, or the straw itself is an open conduit to the

contents of the container, both of which are potential areas of leakage.

(5) Aspects herein are generally directed to a lid structure for a beverage container having a body portion including a straw connector that is configured to disengageably attach to a disengageable straw. Among other things, the disengageable straw facilitates the thorough cleaning of the lid structure, and also provides for a neatly storable container that can fit all of its parts within the interior volume defined by the base structure and the lid structure.

(6) The straw mechanism of the lid structure comprises a spout portion having a spout opening and a frame portion. In an embodiment, the spout portion may be rigid and formed from a hard polymer, a plastic material, a metal, or other rigid material, by way of non-limiting example. In another embodiment, the spout portion may be flexible (i.e., bendable, stretchable, elastic, deformable) and formed from a suitable material such as silicone or other rubber polymer material, by way of non-limiting example. The frame portion frames the spout portion so that the spout portion extends outwardly through an opening of the frame portion and a first surface of the frame portion. Further, the first surface of the frame portion comprises a first indicator that protrudes from the first surface. A spout base portion is wider than the spout portion and sits on a second surface of the frame portion, sealing the opening of the frame portion. The spout opening of the spout portion extends through to the spout base portion. In some aspects the spout portion is shaped like a conventional straw having a cylindrical tubular structure having a circular cross-section with a circular opening. In other aspects, the spout portion may have a tubular structure having any desired cross-sectional shape with a correspondingly shaped opening such as triangular, star, square, etc., by way of non-limiting example. In other aspects, the opening may always be half shape regardless of the cross-sectional shape of the spout portion (e.g., half circle, half star, half triangle, half square, etc., by way of non-limiting example). However, in a preferred embodiment, the spout base portion may always have a half shape opening. In other words, if the spout has a circular shaped opening, the spout base portion may have a half circle shaped opening.

(7) The second surface of the frame portion further comprises a first ring shaped magnet housing protruding from the second surface that houses a ring shaped magnet. Further, an outer surface of the ring shaped magnet housing forms a key with a first ridge and a second ridge, where each partially extend around a circumference of the magnet housing. The first ridge has a first length that is different from the second length of the second ridge. The edge of second surface of the frame portion further comprises a ridge aligning with the first indicator on the first surface of the frame portion and a notch located opposite the ridge.

(8) The engagement cavity of the body portion of the lid structure is configured to engage with the straw mechanism by receiving the magnet housing so that the first surface of the frame portion of the straw mechanism is flush with an exterior surface of the lid structure when engaged to the lid structure. The body portion of the lid structure forms a second ring shaped magnet housing protruding from an inner surface of the body portion that houses a second ring shaped magnet having an opposite polarity to the first ring magnet, creating a strong attraction between the frame portion of the straw mechanism and the body portion of the lid structure. The engagement cavity comprises a first lip and a second lip, the first lip and the second lip form a key hole in the body portion so that the straw mechanism is always received by the body portion in a predetermined configuration. Further, the body portion comprises a couple of stop tabs to lock rotation of the straw mechanism along an 180° path. The second ring shaped magnet is contained within the second ring shaped magnet housing by a cover portion. The cover portion having a half-circle opening configured to align with the half circle opening of the spout base portion. Such that when the half-circle opening of the spout base portion and the half-circle opening of the cover portion are aligned, there is flow from the spout portion and when the half-circle opening and the cover portion and the half-circle opening of the spout portion are misaligned (i.e., on opposite sides), there is no flow from the spout portion.

(9) Aspects herein are generally further directed to a beverage container, the beverage container

being comprised of a base structure and a lid structure having a magnetic straw mechanism. The base structure may be comprised of a double wall plastic structure, glass, or polymer based material structure, or a thermally insulating structure comprised of stainless steel, aluminum, copper, or any other suitable material or combination of materials suitable for insulating the contents of the beverage container from external temperature fluctuations for extended periods of time. The base structure in accordance with aspects herein includes an outer wall and an inner wall. The inner wall of the base structure includes a threaded portion proximal to the opening edge of the base structure. A step edge is provided in the inner wall of the base structure that defines the end of the threaded portion and the start of a non-threaded portion of the inner wall, where the non-threaded portion of the inner wall extends inward toward a relative center of the container further than the internal threaded section of the base structure. The inner wall of may optionally include volume markings configured to signal a volume quantity contained within the base structure as the base structure is filled with a beverage. The volume of the base structure may be sized and shaped to hold between 14 oz. and 100 oz. of a beverage.

(10) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) Illustrative embodiments of the present invention are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein and wherein:

(2) FIG. 1 depicts a perspective view of an assembled beverage system with a lid structure, in accordance with aspects herein;

(3) FIG. 2 depicts a different perspective view of the beverage system shown in FIG. 1 in a disassembled state, in accordance with aspects herein;

(4) FIG. 3 depicts a perspective view of the beverage system in FIG. 1, in a disassembled state;

(5) FIG. 4 depicts a close-up perspective view of the lid structure shown in FIG. 3, in accordance with aspects herein;

(6) FIG. 5A depicts a perspective view of a spout portion of the straw mechanism of the beverage system, in accordance with aspects herein;

(7) FIG. 5B depicts a different perspective view of a spout portion of the straw mechanism of the beverage system, in accordance with aspects herein;

(8) FIG. 5C depicts a different perspective view of a spout portion of the straw mechanism of the beverage system, in accordance with aspects herein;

(9) FIGS. 6A and 6B depict different perspective views of the body portion of the lid structure, in accordance with aspects herein;

(10) FIG. 7 depicts a perspective view of an assembled beverage system with a lid structure in an unlocked state, in accordance with aspects herein;

(11) FIG. 8 depicts a cross-sectional view of the assembled beverage system along the line 8-8 in FIG. 7;

(12) FIG. 9 depicts a perspective view of an assembled beverage system with a lid structure in a dispensing state, in accordance with aspects herein;

(13) FIG. 10 depicts a cross-sectional view of the assembled beverage system along the line 10-10 in FIG. 9;

(14) FIG. 11 depicts a perspective view of an assembled beverage system with a lid structure in a closed state, in accordance with aspects herein;

(15) FIG. 12 depicts a cross-sectional view of the assembled beverage system along the line 12-12 in FIG. 11;

(16) FIGS. 13 and 14 depict different perspective views of a disassembled straw system, in accordance with aspects herein; and

(17) FIGS. 15 and 16 depict different perspective views of an assembled straw system, in accordance with aspects herein.

(18) An overview of the features, functions and/or configurations of the components depicted in the various figures will now be presented. It should be appreciated that not all of the features of the components of the figures are necessarily described. Some of these non-discussed features, such as various couplers, etc., as well as discussed features are inherent from the figures themselves. Other non-discussed features may be inherent in component geometry and/or configuration.

DETAILED DESCRIPTION

(19) The subject matter of embodiments of the present invention is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this patent. Rather, the inventor(s) have contemplated that the claimed subject matter might also be embodied in other ways, to include different features or combinations of features similar to the ones described in this document, in conjunction with other present or future technologies. Further, it should be appreciated that the figures do not necessarily represent an all-inclusive representation of the embodiments herein and may have various components hidden to aid in the written description thereof.

(20) At a high level, a beverage system **100** as shown in FIG. 1 is disclosed. The beverage system **100** comprises a lid structure **102** and a base structure **104** as the main components. The beverage system may optionally further include a handle **106** to facilitate carrying of the beverage system **100**. As further shown in FIG. 1, the lid structure **102** can have a body portion **108**, a straw mechanism **110** having a spout portion **112** with a spout opening **118** through which a user is able to drink fluids contained within the beverage system **100**. The straw mechanism **110** further includes a frame portion **114**. The spout portion **112** extends through an opening **116** of the frame portion **114** so that it extends through and above a first surface (exterior surface) of the frame portion **114**. The frame portion can also comprise a first indicator **120** that guides the user to different positions or states of the straw mechanism **100**, such as, a dispensing state (shown in FIGS. 9 and 10), a locked state or spill-proof state (shown in FIGS. 11 and 12, and a disassemble or cleaning state (shown in FIGS. 7 and 8).

(21) The body portion **108** of the lid structure **102** can comprise a second indicator **122** that is complementary to the first indicator **120**, that when aligned with each other, indicate, for example, that the straw mechanism **110** is in its dispensing state. Further, the body portion **108** comprises a vent **124** for allowing proper air circulation within the beverage system **100** as the contents are pulled from the base structure **104** when the user is drinking from the spout portion **112**, thereby, insuring a proper flow or maintenance of a proper flow of the liquid through the spout opening **118** of the spout portion **112**. As shown, the base structure **104** of the beverage system **100** may optionally include a handle **106** to facilitate transport of the beverage system **100** by a user. Furthermore, as shown in FIG. 2, the beverage system **100** may further include an anti-slip pad **200** on the base or bottom of the base structure **104** to aid in keeping the beverage system **100** from sliding across a flat surface such as a counter, a table, and the like. Exemplary materials that can be used for the anti-slip pad **200** can include, for example, natural rubber, synthetic rubber, rubberized silicone, thermoplastic polyurethane, polyurethane, neoprene, nitrile, and any other suitable material having desirable properties for the anti-slip pad **200**.

(22) FIG. 3 depicts a partially disassembled beverage system **100**. As shown, the body portion **108** and the straw mechanism **110** are able to be separated to facilitate cleaning of the components of the beverage system **100** and in particular, of the lid structure **102**. As shown, the body portion **108** comprises a recess or engagement cavity **308** to accommodate components of the straw mechanism

110 so that the outer surface of the frame portion **114** and the outer surface of the body portion **108** are flush with each other when assembled, as previously shown in FIG. 1. Further, as shown, straw **302** having a straw handle **314** is shown as connected to the body portion **108**. As better seen in FIGS. 4, 5A and 5B, the straw mechanism **110** has a magnetic connection where the second surface of the frame portion **114** has a first magnet housing **318** that protrudes from its inner or second surface, and houses a first magnet **802** (shown in FIG. 8). The first magnet **802** can be ring shaped extending within and around the circumference of first magnet housing **318**, which may also be ring shaped. As shown in FIG. 4, the spout base portion **410** located within a cavity formed in the center of the first magnet housing **318**, comprises a half-circle shaped opening **412**. The second surface (i.e., interior surface) of the frame portion **114** further comprises a projection **402** aligned directly opposite to the first indicator **120** on the first surface of the frame portion. Directly opposite to the projection **402**, the frame portion **114** comprises a notch **400** on the second surface of the frame portion **114**. As will become more apparent in FIGS. 5A-5C and 9-12, when the straw mechanism **110** is in a dispensing state, the notch **400** aligns with the vent seal **300**, and when the straw mechanism **110** is in a sealed state, the projection **402** is aligned with the vent seal **300** to seal the vent **124** by essentially plugging the opening of the vent seal **300**.

(23) The volume of the base structure **104** may be sized and shaped to hold between 14 oz. and 100 oz. of a liquid beverage. The base structure **104** may be comprised of a double wall plastic structure, glass, or polymer based material structure, or a thermally insulating structure comprised of stainless steel, aluminum, copper, or any other suitable material or combination of materials suitable for insulating the contents of the beverage container from external temperature fluctuations for extended periods of time. The base structure **104** comprises an inner wall **330** and an outer wall **332**. An inner diameter of the inner wall **330** is generally smaller than an outer diameter of the outer wall **332**, thereby providing a thickness to the base structure **104** between the inner wall **330** and the outer wall **332**. The thickness may aid in providing insulation properties to the base structure **104** to prevent rapid cooling, rapid warming and/or sweating when containing cold or iced liquid beverages. Further, in order to improve gripping and/or gripping comfort, the base structure **104** may optionally, further comprise handle **106** which may or may not be provided with an additional frictional surface that is coated or otherwise covered with a rubber, such as for example, natural rubber, synthetic rubber, rubberized silicone, thermoplastic polyurethane, polyurethane, neoprene, nitrile, and any other suitable material having desirable properties.

(24) As also shown in FIG. 4, the frame portion **114** further comprises a first ridge **404** and a second ridge **406** that project from the first magnet housing **318** to form the “key” portion of the straw mechanism **110**, and the engagement cavity **308** forms the “lock” portion of the lid structure by having an opening defined by a first lip extension **304** and a second lip extension **306** or conversely a first lip notch **322** and a second lip notch **324**. The first ridge **404** has a first length and the second ridge **406** has a second length that is different from the first length. The first lip notch **322** also has the first length and the second lip notch **324** also has the second length. As such, the straw mechanism **110** can fit into the engagement cavity **308** in only one predetermined way (i.e., the first ridge **404** passes through the first lip notch **322** and the second ridge **406** passes through the second lip notch **324**), as will become more apparent from the remaining figures.

(25) As further shown in FIGS. 3, 4, 6A and 6B, the vent seal **300** may be made from a flexible rubber or silicone material having an opening through which air is able to flow when not sealed. The first lip extension **304** and the second lip extension **306** are further equipped with stop tabs **320** respectively, that are configured to limit the rotation of the straw mechanism **110** with respect to the body portion **108** to 180° (i.e., half circle path). The body portion **108** further comprises threading **328** that secure the lid structure **102** to the base structure **104**. The base structure comprises threading **326** on its interior wall **330** defining the interior volume **316** that is complementary to the threading **328**. As shown, the base structure **104** has an interior volume **316** where a beverage can be contained in the base structure **104**.

(26) As will become more apparent in FIGS. 8, 10, and 12, the body portion **108** houses a second magnet **804** within the engagement cavity **308** that is secured and covered by a cover portion **310**. The cover portion **310** comprises a second half-circle shaped opening **312** that works with the half-circle shaped opening **412** of the spout base portion **410** to form the straw mechanism **110**. The straw mechanism **100** is able to transition from a dispensing state and a closed state (sealed state). The straw mechanism **100** is in the dispensing state when the half-circle shaped opening **312** and the half-circle shaped opening **412** are aligned with each other. The closed state, on the other hand, is when the half-circle shaped opening **412** and the half-circle shaped opening **312** are not aligned with each other (are directly opposite to each other).

(27) FIG. 7 shows the beverage system **100** in an unlocked state **700**. When in the unlocked state **700**, the first indicator **120** is located a quarter turn of the straw mechanism **110** between the second indicator **122** and the vent **124**, or in other words, at a 90° clockwise rotation point from the second indicator **122**. As better shown in FIG. 8, the first ridge **404** is aligned with the first lip notch **322** and the second ridge **406** is aligned with the second lip notch **324** such that the straw mechanism **110** can be pulled out or separated from the body portion **108** of the lid structure **102**, as shown previously in FIGS. 3 and 4. The removal of the straw mechanism **110** from the body portion **108** facilitates access to areas of the lid structure **102** underlying the straw mechanism **110**, for cleaning, by way of example. As described above and further shown in the cross-sectional view **800** shown in FIG. 8, the first magnet housing **318** houses a first magnet **802**, which when the straw mechanism **110** is within the engagement cavity **308**, the first magnet **802** is aligned with a second magnet **804** housed within the engagement cavity **308** and secured in place by the cover portion **310**. The first magnet **802** and the second magnet **804** have opposing polarities so that they strongly attract each other and are held magnetically even in this unlocked state **700** so that the user has better control of the parts and the parts will not just come loose on their own. The user has to make a pulling motion on the straw mechanism **110** to pull it apart from the body portion **108**.

(28) As shown, in FIG. 8, the first magnet **802** is secured within the first magnet housing **318** by elastic gasket **806**. Further, the cover portion **310** is sealed by elastic seal **808** and elastic seal **810**, thereby effectively securing the second magnet **804** within the body portion **108**. As shown, when the straw mechanism **110** is within the engagement cavity **308**, the first magnet **802** and the second magnet **804** are in alignment with each other. It is contemplated in accordance with aspects herein that the first magnet **802** and the second magnet **804** are each one piece ring shaped magnets. In accordance with other aspects, rather than being ring shaped, the first magnet **802** and the second magnet **804** may be open ring or in other words, arch shaped. The first arch shaped magnet may occupy at least 50% of the first magnet housing **318** and the second magnet **804** may also be arch shaped to match the shape of the first magnet **802**. Alternatively, the arch shaped magnets may be smaller, each occupying, for example, less than 50% of the first magnet housing **318**. In such embodiments, there may be two or more arch shaped magnets distributed at different locations within the first magnet housing **318**, each occupying at least 10% of the first magnet housing **318**. The arch shaped magnets within the engagement cavity **308** may then be made to match up the arch shaped magnets within the first magnet housing **318**. As further shown, when the straw **302** is connected to the body portion **108**, a straw fitting **1402** of the body portion **108** is configured to fit within the straw opening **814**, which is located within straw connection port **1302**. The straw connection portion **1302** defines a socket **1306**, as will become more apparent in FIG. 13. The straw **302** and the body portion **108** are sealed by elastic gasket **812** when connected to each other.

(29) FIG. 9 shows the beverage system **100** in a locked and dispensing state **900**. When in the locked and dispensing state **900**, the first indicator **120** is rotated counter-clockwise from the disassemblable state **700** as shown by arrow **902**, to align the first indicator **120** and the second indicator **122**. In other words, the straw mechanism **110** is rotated a quarter turn counter-clockwise from the unlocked state so that the first indicator **120** is positioned at a 0° rotation point from the second indicator **122**. As better shown in cross-sectional view **1000** in FIG. 10, the first ridge **404** is

aligned with the second lip extension **306** and the second ridge **406** is aligned with the first lip extension **304** such that the straw mechanism **110** is locked within the engagement cavity **308** and thus, cannot be separated from the body portion **108** of the lid structure **102**. In the locked and dispensing state **900**, a connected opening **1006** is formed where the spout opening **118**, the half-circle shaped opening **412**, the half-circle shaped opening **312**, and the straw opening **814** are all aligned with each other, thereby, allowing free flow of the contents within the interior volume **316** through the connected opening **1006**. As the contents are being emptied, air can be pulled in from the vent **124**, which has a vent seal **300** with a vent opening **301**. This prevents a vacuum from being formed within the interior volume **316**, which would make the contents harder and harder to pull through the spout portion **112**. As shown, the notch **400** of the frame portion **114** is aligned with the vent seal **300** so as to not impede air flow through the vent opening **301**.

(30) FIG. **11** shows the beverage system **100** in a locked and non-dispensing state **1100**. When in the locked and non-dispensing state **1100**, the first indicator **120** is rotated clockwise from the disassemblable state **700** as shown by arrow **1102**, to place the first indicator **120** and the second indicator **122** on opposite sides from one another, or in other words, at a 180° rotation point from the second indicator **122**. As better shown in cross-sectional view **1200** of FIG. **12**, the first ridge **404** is aligned with the first lip extension **304** and the second ridge **406** is aligned with the second lip extension **306** such that the straw mechanism **110** is also locked within the engagement cavity **308** and thus, cannot be separated from the body portion **108** of the lid structure **102**. In the locked and non-dispensing state **1100**, a disconnected opening **1202** is formed where the half-circle shaped opening **412** and half-circle shaped opening **312** are misaligned with each other (i.e., on opposite sides of each other), thereby blocking flow from the straw opening **814** into the spout opening **118**. Additionally, the projection **402** on the underside of the frame portion **114** is aligned with the vent seal **300**, thereby blocking the vent opening **301** of the vent seal **300**, resulting in a blocked opening **1204**, thereby, effectively creating a leak-proof configuration.

(31) FIG. **13** shows the straw assembly view **1300**. As shown, the straw **302** has a straw handle **314** with a straw tab **1308**. The straw **302** may be made from any suitable material that is safe for use such as stainless steel, glass, silicone, bisphenol A (BPA) free plastics, and the like. The handle **314** may be molded onto the straw **302**, or may be made as one piece with the straw **302**, or maybe fitted onto straw **302**. The handle **314** may be made from the same material as the straw **302**, or a different material as the straw **302**. The handle **314** may be removable from straw **302** or permanently affixed to straw **302**. As shown, the body portion **108** of lid structure **102** has a straw connection port **1302** on the underside of the lid structure **102**. The straw connection port **1302** can be located opposite the engagement cavity **308**, and defines a socket **1306**. The straw connection port **1302** has a vertical guide **1310** (parallel to a direction of the straw length) that transitions into a horizontal guide **1304** that is perpendicular or orthogonal to the vertical guide **1310**. Further, as shown in FIG. **14** with the body of the connection port **1302** removed, a straw fitting **1402** within the socket **1306** is configured to fit within the straw opening **814**. At the base of the straw fitting **1402** is an elastic gasket **812** to seal the fitting of the straw **302** and the body portion **108**.

(32) As shown in FIGS. **15-16**, the steps for assembling the straw **302** to the lid structure **102** are depicted. Looking at FIG. **15**, the tab **1308** is glided upwards in a vertical direction **1502** through the vertical guide **1310** of the straw connection portion **1302**. To facilitate the process, the straw handle **314** may be directly aligned with the tab **1308**. Once the straw is within the socket **1306**, FIG. **16** illustrates that the straw **302** can be turned in a clockwise direction **1602**, guiding the tab **1308** through the horizontal guide **1304**, thereby locking the straw to the body portion **108**. Again, this motion can be facilitated by placing a clockwise directional force on the straw handle **314**.

(33) Additionally, although some exemplary implementations of the embodiments described herein are shown in the accompanying figures, these implementations are not intended to be limiting. Rather, it should be understood that the various embodiments and aspects described herein may be implemented upon any insulated container.

(34) Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of the present invention. Embodiments of the present invention have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from the scope of the present invention.

Claims

1. A lid structure for a beverage container, comprising: a body portion having a top side that defines an engagement cavity sized to receive a straw mechanism, and a first opening offset from a central axis of the body portion, the first opening being in fluid communication with an interior volume of the beverage container; and a straw mechanism configured to be removably inserted into the engagement cavity, the straw mechanism including a bottom portion having at least two protruding ridges, each protruding ridge being sized and arranged to pass through a corresponding notch formed on or within the engagement cavity so that the straw mechanism can be fully inserted only in a single orientation, and a spout portion having a distal end surface that presents a second opening offset from the central axis, wherein the straw mechanism is rotatable within the engagement cavity between an open state in which the second opening is aligned with the first opening so that fluid may pass from the container interior to the spout portion, and a closed state in which the first opening is blocked to inhibit fluid flow.
2. The lid structure of claim 1, the at least two protruding ridges being of different sizes.
3. The lid structure of claim 1, wherein the second opening and the first opening are configured such that, in the closed state, the first opening is covered or occluded by a portion of the straw mechanism.
4. The lid structure of claim 1, the body portion having one or more stop elements that limit rotation of the straw mechanism within the engagement cavity to about 180 degrees between the open state and the closed state.
5. The lid structure of claim 1, wherein the straw mechanism includes a frame portion that is generally flush with an exterior surface of the body portion when inserted, the frame portion supporting the spout portion and the bottom portion.
6. The lid structure of claim 5, wherein the frame portion presents at least one indicator on an upper surface and the body portion presents a complementary indicator, such that alignment of the indicators denotes whether the straw mechanism is in the open state or the closed state.
7. The lid structure of claim 1, further comprising a vent formed in the body portion and configured so that the straw mechanism, upon rotation, selectively seals or unseals the vent.
8. The lid structure of claim 1, further comprising a detachable straw removably coupled to an underside of the body portion, the detachable straw extending into the interior volume of the beverage container.
9. The lid structure of claim 1, wherein the straw mechanism and the body portion each include a magnet of opposite polarity, producing a magnetic bias that retains the straw mechanism in the engagement cavity.
10. The lid structure of claim 1, wherein the engagement cavity defines a keyhole opening with at least two differently sized notches, and wherein the bottom portion of the straw mechanism includes the at least two protruding ridges, each protruding ridge being sized and arranged to pass through a corresponding one of the differently sized notches of the keyhole opening.
11. A beverage container comprising: a base structure including a threaded region proximate an upper opening; and a lid structure including a body portion having threads complementary to the threaded region of the base structure to removably secure the lid structure onto the base structure, a top side that defines an engagement cavity sized to receive a straw mechanism, and a first opening

offset from a central axis of the body portion, the first opening being in fluid communication with an interior volume of the beverage container; and a straw mechanism configured to be removably inserted into the engagement cavity, the straw mechanism having a bottom portion having at least two protruding ridges, each protruding ridge being sized and arranged to pass through a corresponding notch formed on or within the engagement cavity so that the straw mechanism can be fully inserted only in a single orientation, and a spout portion having a distal end surface that presents a second opening offset from the central axis, wherein the straw mechanism is rotatable within the engagement cavity between an open state in which the second opening is aligned with the first opening so that fluid may pass from the container interior to the spout portion, and a closed state in which the first opening is blocked to inhibit fluid flow.

12. The beverage system of claim 11, wherein the at least two protruding ridges are of different sizes.

13. The beverage system of claim 11, wherein the second opening and the first opening are configured such that, in the closed state, the first opening is covered or occluded by a portion of the straw mechanism.

14. The beverage system of claim 11, the body portion having one or more stop elements that limit rotation of the straw mechanism within the engagement cavity to about 180 degrees between the open state and the closed state.

15. The beverage system of claim 11, wherein the straw mechanism includes a frame portion that is generally flush with an exterior surface of the body portion when inserted, the frame portion supporting the spout portion and the bottom portion.

16. The beverage system of claim 15, wherein the frame portion presents at least one indicator on an upper surface and the body portion presents a complementary indicator, such that alignment of the indicators denotes whether the straw mechanism is in the open state or the closed state.

17. The beverage system of claim 11, further comprising a vent formed in the body portion and configured so that the straw mechanism, upon rotation, selectively seals or unseals the vent.

18. The beverage system of claim 11, further comprising a detachable straw removably coupled to an underside of the body portion, the detachable straw extending into the interior volume of the beverage container.

19. The beverage system of claim 11, wherein the straw mechanism and the body portion each include a magnet of opposite polarity, producing a magnetic bias that retains the straw mechanism in the engagement cavity.

20. The beverage system of claim 11, wherein the engagement cavity defines a keyhole opening with at least two differently sized notches, and wherein the bottom portion of the straw mechanism includes the at least two protruding ridges, each protruding ridge being sized and arranged to pass through a corresponding one of the differently sized notches of the keyhole opening.
